

CSC 273.2

Introduction to Cryptography and Blockchain Technology

Study Notes — Godswill | UNIPORT, Computer Science

Chapter Overview

This course introduces the technology underlying cryptocurrencies and other decentralized digital systems. It starts with the blockchain itself — the distributed ledger that makes trustless, peer-to-peer value transfer possible — then builds up through cryptocurrencies and native coins, the broader universe of tokens (fungible and non-fungible), how wallets and addresses actually work, and the consensus mechanisms (mining and staking) that keep these networks secure without a central authority. From there, it covers how people actually trade and use these assets (exchanges, DeFi), how real-world and digital assets alike are being tokenized, the major blockchain networks in use today, the cryptographic foundations (history, algorithms, and implementations) that make all of this secure, how the crypto market is measured and ranked, the programming tools used to build on these platforms, and finally real-world applications, case studies, and the regulatory outlook shaping where this technology is headed.

1. Foundations of Blockchain Technology

1.1 What Is a Blockchain?

A blockchain is a peer-to-peer computer network running a distributed ledger — consisting of nodes and edges — where data is securely chained together chronologically using cryptography. It lets a digital ledger be maintained securely, without a middleman, by a network rather than a single central authority.

1.2 Nodes

Nodes are powerful computers spread around the world that maintain and manage the blockchain. Each node uses advanced encryption algorithms to secure data, and can run special code known as smart contracts — self-executing code stored directly on the blockchain that carries out an agreement automatically when its conditions are met. Because many independent nodes each hold a copy of the same ledger, the system achieves what is sometimes called programmable trust: participants don't need to trust a central authority, only the shared rules encoded into the network itself.

1.3 Public vs Private Blockchains

Type	Description	Examples
Public Blockchain	Open to anyone; anyone can join as a node, view the ledger, and transact.	Bitcoin, Ethereum
Private / Permissioned Blockchain	Access is restricted to approved participants, often used by institutions.	Enterprise/consortium ledgers
CBDC (Central Bank Digital Currency)	A digital currency issued and controlled directly by a country's central bank/government, distinct from public, decentralized cryptocurrencies.	Nigeria's eNaira

Note the key distinction: public cryptocurrencies (like Bitcoin, Ethereum, Solana, or PulseChain) are decentralized and not issued by any government, whereas CBDCs are centrally issued and controlled — they use similar underlying digital-ledger ideas but represent a fundamentally different model of control.

2. Cryptocurrencies, Coins, and Native Assets

2.1 What Is Cryptocurrency?

Cryptocurrency is a digital and recorded virtual/issued form of currency, secured by cryptography, that exists through a blockchain. It can be public, private, or hybrid. Cryptography is what secures cryptocurrency — it's the reason ownership and transactions can be verified without a central authority.

2.2 Coins vs Tokens — The Core Distinction

This is one of the most important conceptual distinctions in the field:

	Coin	Token
Definition	An asset operating on its own independent, standalone blockchain, acting as that network's native gas/currency.	A programmable digital asset generated using smart contract code, living on a host blockchain rather than its own.
Blockchain	Has its own dedicated blockchain.	Lives on someone else's existing blockchain (e.g. Shiba Inu or Chainlink on Ethereum).
Examples	Bitcoin (Bitcoin blockchain), Ether (Ethereum blockchain)	Most utility, governance, and DeFi assets

Coins pay the network's own gas fees and are what the underlying blockchain runs on; tokens, by contrast, are highly programmable digital assets that piggyback on a host blockchain's infrastructure via smart contracts.

2.3 Central Bank Digital Currency (CBDC)

A CBDC is actually issued and controlled by a government/central bank — unlike Bitcoin or Ethereum, which are decentralized and issued by no single authority. Nigeria's eNaira is a real-world example of a CBDC.

3. Tokens: Types and Characteristics

Tokens are programmable digital assets. They are highly programmable and generated using smart contract code on a host blockchain. Since the concept of tokens was pioneered on Ethereum (Ethereum had the first token), most token types trace back to its smart contract standards.

3.1 Major Types of Tokens

Token Type	What It Does	Example
Utility Tokens	Grant access to a specific product, service, or functionality within a blockchain ecosystem.	Uniswap

Token Type	What It Does	Example
Governance Tokens	Give holders the right to vote on decisions, proposals, and updates influencing a blockchain project's future — often used in a Decentralized Autonomous Organization (DAO).	DAO governance tokens
Tokenized Stocks	Give exposure to (or full control of) a real stock, letting holders gain that exposure without buying the stock directly through a traditional broker.	Tesla token, Nvidia token
Stablecoins	Coins/tokens pegged to a stable external asset, most commonly the US Dollar.	USDT, DAI
DeFi Tokens	Tokens native to decentralized finance applications, often used for lending, trading, or earning interest.	Hex, YFI
NFTs (Non-Fungible Tokens)	Unique, non-interchangeable digital tokens representing ownership of a specific item or asset.	Digital art, collectibles

3.2 Decentralized Autonomous Organizations (DAOs)

A DAO (Decentralized Autonomous Organization) is a blockchain-based organization governed by its governance token holders rather than a traditional management hierarchy. Votes on proposals are typically weighted by the number of tokens held — whoever holds the largest number of governance tokens carries the most voting power.

4. Fungible vs Non-Fungible Tokens

Fungible Tokens (FT): Interchangeable, identical-value tokens — one unit is exactly equivalent to another unit of the same token (comparable to how one dollar bill is interchangeable with another).

Non-Fungible Tokens (NFT): Unique tokens where each one is distinct and not directly interchangeable with another, typically representing ownership of a specific, one-of-a-kind digital or real-world item.

4.1 Relevant Ethereum Token Standards

Standard	Purpose
ERC-20	The standard for fungible tokens on Ethereum — the most widely used token standard for currencies, utility tokens, and governance tokens.
ERC-721	The standard for non-fungible tokens (NFTs) on Ethereum, where each token ID is unique.

The Ethereum Virtual Machine (EVM) is the runtime environment that executes smart contracts on Ethereum (and EVM-compatible chains), making standards like ERC-20 and ERC-721 possible.

5. Blockchain Addresses, Key Holders, and Wallets

A wallet is software or hardware storing the cryptographic keys for your assets. Holding cryptocurrency doesn't mean owning a physical file of money — it means holding the cryptographic keys that let you transfer

digital ledger records on the blockchain. A wallet acts like a key ring: it stores the private keys needed to unlock and access your coins on the blockchain, not the coins themselves (which live on the ledger).

5.1 What Wallets Do

- Confirm your identity to the network.
- Grant you the required access to authorize transactions involving your assets.
- Store your cryptographic keys securely — software wallets (apps) or hardware wallets (dedicated physical devices).

Common examples include MetaMask (a popular software/browser wallet) and dedicated hardware wallets, which store private keys offline for extra security.

6. Consensus Mechanisms: Mining and Staking

6.1 What Is Consensus?

Consensus follows a 'majority decides' mentality: blockchain technology relies on the majority of participating (decisive) nodes agreeing on the final decision/state of the ledger. This is the core principle guiding blockchain technology — it's how a decentralized network of mutually distrusting nodes can still agree on a single, shared version of the truth.

6.2 Proof of Work and Mining

Proof of Work is a consensus mechanism where participants (miners) compete to solve a math equation; whoever solves it first gets to add the next block and receives a reward — the network's native gas/coin (e.g. Bitcoin's own gas is paid out in Bitcoin). Mining is the process of solving these math problems to generate new coins and validate the ledger.

6.3 Mining vs Staking

Mechanism	How It Works
Mining (Proof of Work)	Participants use computational power to solve math problems; the first to solve it validates the next block and earns a reward in the network's native coin.
Staking (Proof of Stake)	Participants lock up (stake) an amount of the native coin/token as collateral to be selected to validate transactions, earning rewards without heavy computational competition.

Both mining and staking function as self-sustaining loops: they secure the network while simultaneously rewarding the participants who do the securing.

7. Exchanges: Centralized vs Distributed

Exchange Type	Description	Examples
Centralized Exchanges (CEX)	A company-operated platform that holds custody of users' funds and matches buyers/sellers internally.	Major mainstream exchanges
Distributed / Decentralized Exchanges (DEX)	Peer-to-peer trading directly on the blockchain via smart contracts, with no central custodian holding user funds.	Uniswap

Because DEX activity checks directly on-chain, it is directly linked to the blockchain itself — this transparency is part of why tools that analyze DEX data (see Section 12) are considered especially reliable indicators of genuine trading activity.

8. Decentralized Finance (DeFi)

DeFi (Decentralized/Distributed Finance) refers to financial applications — lending, trading, earning interest, and more — built on blockchain infrastructure without traditional banks or intermediaries. DeFi tokens (like Hex or YFI) are native to these applications, and Decentralized Finance Tokens broadly represent this shift toward modern finance that can pay holders interest directly through code rather than a bank.

9. Asset Tokenization

9.1 What Is Tokenization?

Tokenization is the process of converting an asset — whether a digital asset (like a cryptocurrency) or a traditional real-world asset (real estate, stocks, commodities, or fiat currencies) — into a digital token on a blockchain network. The asset being tokenized can be a physical asset.

9.2 The Tokenization Process

- **Asset Selection:** choosing what is being tokenized — it can be a physical asset (real estate, precious metals) or a purely digital one.
- **Smart Contract Creation:** writing the smart contract that will represent and manage the token (e.g. using Solidity for Ethereum-based tokens).
- **Minting:** the act of creating (issuing) the actual tokens on the blockchain according to the smart contract's rules.
- **Listing/Trading:** once minted, a token can be listed on an exchange for trading.

A smart contract for a token is written inside an already-existing blockchain — if a chain isn't ready (i.e. still being written/developed in some programming language), it isn't yet possible to release a coin on it. This is why most tokens rely on established host blockchains (like Ethereum) rather than launching an entirely new blockchain from scratch.

9.3 Relevant Standards by Network

Network	Smart Contract Language	Fungible Token Standard
Ethereum	Solidity	ERC-20
PulseChain	Solidity (EVM-compatible)	Similar EVM-based standard

9.4 Examples of Tokenized Assets

- Commodity Tokenization: precious metals such as gold and silver represented as tokens.
- Real Estate Tokens: property ownership/value represented as tradeable digital tokens.
- Tokenized Stocks: stock exposure (e.g. Tesla, Nvidia) made available via tokens, without needing to buy through a traditional broker.
- Blockchain 'Meme' Tokenization: novelty/community-driven coins (e.g. a themed coin created around a cat, a teddy bear, or even a public figure) tokenized largely for cultural/speculative purposes rather than utility.

A basic market dynamic worth noting: the lower the number of holders willing to sell when a coin first begins trading, the higher the price of the coin tends to be pushed, since available supply is scarce relative to demand.

10. Major Blockchain Networks and Their Ecosystems

10.1 Bitcoin and Its Forks

Bitcoin, the first and most dominant cryptocurrency, has inspired several related networks, some created directly as 'forks' (splits) of its codebase/history:

- Bitcoin (BTC) — the original.
- Bitcoin SV (BSV) and Bitcoin Cash (BCH) — forks of Bitcoin, created by community disagreements over how the network should scale/evolve.
- Dogecoin (DOGE) — originally created as a lighthearted, meme-based cryptocurrency, later gaining major real-world attention partly due to high-profile investment interest (e.g. Elon Musk).
- Litecoin (LTC) — another Bitcoin-inspired network, often described as designed for faster, lighter transactions.

10.2 Ethereum and Its Ecosystem

Ethereum introduced programmable smart contracts and the first tokens, making it the primary host blockchain for a huge share of all tokens in existence. Binance operates its own related networks:

- BNB Chain — Binance's own native blockchain.
- BSC (Binance Smart Chain) — an EVM-compatible chain associated with Binance, allowing Ethereum-style smart contracts and tokens.

10.3 Other Notable Networks

- Solana — known for high transaction throughput and low fees.
- PulseChain — an EVM-compatible network, using a fungible token standard similar to Ethereum's.

10.4 A Historic Milestone: The First Real-World Crypto Purchase

Before Bitcoin gained real-world purchasing recognition, cryptocurrencies were sometimes dismissively referred to as 'experimental coins,' obtained only via mining. A defining historical moment came when someone famously spent a large amount of Bitcoin to purchase everyday goods (widely remembered as the 'Bitcoin Pizza' event) — this transaction is remembered as initiating the first real-world purchase made with cryptocurrency, marking the moment Bitcoin's value moved from purely theoretical to practically usable.

11. Cryptography Fundamentals

11.1 History of Cryptography

Cryptography — the practice of securing information so only intended parties can access it — long predates computers. Early cryptography relied on simple substitution ciphers (such as the Caesar cipher used in ancient Rome) to hide messages during wartime and diplomacy. The 20th century brought mechanical and electromechanical cipher devices (such as the Enigma machine), and the rise of computing in the mid-to-late 20th century enabled far more sophisticated mathematical cryptography — including the public-key cryptography breakthroughs of the 1970s that made modern secure digital communication, and eventually blockchain technology, possible.

11.2 Types of Cryptographic Implementation

Type	How It Works	Blockchain Use
Symmetric Cryptography	The same key is used to both encrypt and decrypt data.	Less common directly in blockchain due to key-sharing challenges.
Asymmetric (Public-Key) Cryptography	Uses a mathematically linked key pair — a public key (shareable) and a private key (kept secret) — where data encrypted with one can only be decrypted with the other.	The basis for wallet addresses (public key) and transaction signing (private key).
Cryptographic Hashing	A one-way function that converts input data into a fixed-length output ('hash'); even a tiny input change drastically changes the output, and the process cannot be reversed.	Used to link blocks together securely and verify data integrity.

11.3 Common Cryptographic Algorithms

- SHA-256: a widely used cryptographic hash function, notably used in Bitcoin's Proof of Work mining process.
- ECDSA (Elliptic Curve Digital Signature Algorithm): used to generate the public/private key pairs behind most blockchain wallet addresses and to digitally sign transactions.
- RSA: an older but still widely used asymmetric encryption algorithm, foundational to modern public-key cryptography generally.
- AES (Advanced Encryption Standard): a common symmetric encryption algorithm used to secure data at rest (e.g. encrypting a wallet file on a device).

In blockchain specifically, hashing secures and links each block to the one before it (making tampering detectable), while public-key cryptography secures individual wallet addresses and proves that a transaction was authorized by the actual holder of the private key, without ever revealing that private key itself.

12. Crypto Market Analysis and Rankings

12.1 Why Crypto Ranking Exists

Crypto ranking emerged directly out of the proliferation of blockchain projects: with thousands of coins and tokens in existence, rankings help investors evaluate the size, stability, and popularity of tokens, filtering genuine quality from noise. Since crypto value is inherently tied to money, some element of monetary value has to be included in any ranking system — which is exactly what Market Capitalization provides.

12.2 Market Capitalization — The Ranking Engine

Market Capitalization = Current Trading Price × Circulating Supply

This is the primary metric for gauging a coin's size and is used for risk assessment — the coin with the highest market cap value becomes the No. 1 ranked asset. As of recent data, the total global crypto market cap is roughly \$2.15 trillion, with Bitcoin alone commanding around 56–58% of that total (Bitcoin market cap has been cited around \$1.19 trillion, with Ethereum around \$193 billion) — meaning Bitcoin remains the dominant force in the industry.

12.3 Ranking Platforms (Top Dashboards)

Platform	What It's Known For
CoinMarketCap	The industry standard for comprehensive market data — prices, trading volume, and tracking across the global crypto market.
CoinGecko	A popular independent tracking platform, known for tracking developer activity and NFT rankings; tracks over 10,000 assets.
CoinRanking / CryptoSlate / CryptoRank	Alternative, more niche ranking tools.

12.4 Visual Analytics — Tracking with Bubbles

Beyond a traditional (if somewhat flat) list view, several tools visualize complex on-chain data using physics-based 'bubble' visualizations, where a bubble's size corresponds to a coin's market cap or its percentage price change:

- CryptoBubbles & BanterBubbles: interactive visual tools showing market movements and relative size using bubbles.
- Bubblemaps.io: maps wallet connections directly on-chain, helping reveal insider clusters and large holders ('whales') who could move the market.
- DexScreener & GeckoTerminal: real-time analytics platforms tracking decentralized exchange (DEX) liquidity and token trading pairs across 80+ blockchains, using raw, unmoderated on-chain data.

12.5 Key Valuation and Ranking Metrics

Metric	What It Measures	Purpose
Trading Volume	24-hour buy/sell demand across exchanges	Shows real, current market

Metric	What It Measures	Purpose
	(total exchange volume = total buy and sell activity across all exchanges combined).	interest.
Supply Dynamics	Circulating supply vs. Max supply (the maximum number of coins that can ever exist, versus how many are currently available).	Affects scarcity and long-term value.
FDV (Fully Diluted Valuation)	What the market cap would be if the maximum possible supply were all in circulation.	Reveals future dilution risk.
TVL (Total Value Locked)	The total value of assets deposited/locked in a DeFi protocol.	A high TVL suggests genuine usage, not a 'zombie' project.
On-Chain Activity	Active addresses and on-chain transaction volume.	Shows organic adoption, not just speculative price action.
Sentiment & Momentum	The Fear & Greed Index, All-Time-High (ATH) tracking.	Gauges overall market emotion.

12.6 Zombie Coins

A 'Zombie Coin' is a token with a high theoretical market cap based on a massive token supply, but with little to no real-world usage, trading volume, or liquidity behind it. Because market cap alone can be misleadingly inflated by simply minting a huge supply, experienced analysts always cross-check market cap against real usage metrics (trading volume, TVL, active addresses) before trusting a ranking at face value.

12.7 Rank Fixing

Rank fixing refers to overriding or bypassing a ranking website's normal methodology — for example, a large exchange adjusting how a top-ranking coin's rank is displayed, sometimes with the effect (intended or not) of driving away investors from a competing asset. This is a reminder that ranking platforms, while useful, are not infallible and can be influenced.

13. Programming in Cryptography and Blockchain

Different blockchain ecosystems and cryptographic tasks call for different programming languages, each suited to its platform's needs:

Language	Primary Use in This Field
C++	Used to build Bitcoin's core client software — chosen historically for its performance and low-level control.
Solidity	The dominant smart contract language for Ethereum and other EVM-compatible chains (e.g. PulseChain) — used to write the code behind tokens, DeFi protocols, and DAOs.
JavaScript	Widely used for building blockchain-facing web applications, wallets, and interacting with smart contracts from the browser (e.g. via libraries that connect to MetaMask).
Python	Popular for prototyping cryptographic algorithms, blockchain data analysis, and automation/trading scripts due to its readability and rich libraries.
Java	Used in Hyperledger, an enterprise-focused permissioned blockchain framework favored for private/consortium blockchain applications.

14. Real-World Applications, Case Studies, and Global Outlook

14.1 Real-World Applications of Cryptography and Blockchain

- Public Blockchain (Bitcoin): peer-to-peer digital currency, censorship-resistant value storage and transfer.
- Private Blockchain / CBDCs (e.g. Nigeria's eNaira): government-controlled digital currency infrastructure, built on similar underlying ledger technology but with centralized control.
- DeFi applications: lending, trading, and earning interest without traditional banking intermediaries.
- Asset Tokenization: bringing real-world assets like real estate, commodities, and even traditional stocks onto blockchain rails for improved liquidity and accessibility.

14.2 Emerging Markets and Nigeria's Position

Emerging economies — with Nigeria specifically highlighted — are seeing growing interest in tokenization and cryptocurrency adoption, driven by opportunities in cross-border transactions and the tokenization of physical or real-world assets to improve liquidity and accessibility. Cross-border tokenized asset platforms are explored as a way to bridge financial networks between regions like the US and Africa and the rest of the world, making investment opportunities more decentralized and globally accessible.

14.3 Regulatory Outlook (2025–2026)

Regulatory frameworks are actively evolving to catch up with the technology. In the United States, proposed frameworks such as the GENIUS Act and the Clarity Act are discussed as upcoming regulation aimed at driving institutional adoption of tokenized financial assets and providing clearer legal treatment for digital assets generally — a trend likely to shape how both coins and tokenized real-world assets are treated going forward.

Chapter Summary

Putting the whole course together:

- A blockchain is a peer-to-peer, cryptographically secured distributed ledger maintained by nodes, which can be public, private, or a government-controlled CBDC.
- Coins run on their own dedicated blockchain and pay that network's gas; tokens are programmable digital assets built via smart contracts on a host blockchain — spanning utility, governance, tokenized stock, stablecoin, DeFi, and NFT types, and split further into fungible (ERC-20) versus non-fungible (ERC-721) standards.
- Wallets don't store coins directly — they store the cryptographic keys that prove ownership and authorize transactions on the ledger.
- Consensus (via mining/Proof of Work or staking/Proof of Stake) is how a decentralized network of nodes agrees on a single shared ledger state without a central authority.
- Exchanges (centralized or decentralized) and DeFi applications are how these assets are actually traded and put to financial use.
- Tokenization extends blockchain's reach to real-world assets — real estate, commodities, and stocks — not just native digital currencies.
- Cryptography (hashing and public-key cryptography, built from algorithms like SHA-256, ECDSA, and RSA) is the mathematical foundation securing every layer of this system.

- Crypto rankings, led by Market Cap (price × circulating supply) and cross-checked against metrics like TVL and trading volume, help investors separate genuine projects from 'zombie coins.'
- All of this is built using a mix of languages — C++, Solidity, JavaScript, Python, and Java — and is expanding into real-world applications and emerging markets, under an evolving global regulatory landscape.